

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		<u>Earth's</u> magnetic field mentioned			1	1		1
	(b)		$\ln a = -3.22$ (1) 0.000 127 and -8.97 for B and $\ln B$ (1) Data point within $\frac{1}{2}$ square (1) ecf	1	1 1		3	3	3
	(c)		Method for gradient (1) Answer = -0.94 ± 0.01 (no penalty for missing minus) (1)	1	1		2	2	2
	(d)		Taking logs correctly e.g. $\ln B = \ln \left(\frac{\mu_0 I}{2\pi} \right) + \ln \left(\frac{1}{a} \right)$ (1) accept $\ln B = \ln \left(\frac{\mu_0 I}{2\pi a} \right)$ In correct form e.g. $\ln B = -\ln a + \ln \left(\frac{\mu_0 I}{2\pi} \right)$ (1)		2		2	1	2
	(e)		Any 3 × (1) from: -Straight line -Through all error bars or close to points or small scatter -Gradient close to -1 -Intercept -12 agrees with small permeability			3	3	3	3